

Introduction to Microeconomics

Lecture 10—Profit Maximization, Supply and Market Equilibrium in Perfectly Competitive Markets

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ECON101
Fall 2024

Perfectly Competitive Market

A perfectly competitive market is characterized by

- Many small firms compared to the total size of the market.
- Homogeneous product.
- Full information.
- Negligible Transaction Costs.
- Free Entry and Exit.

We will develop the theory of the firm operating in a perfectly competitive market. However, many of the insights that we develop here extends to cases in which some of the characteristics above are not true. Turns out that none of the above are essential for a market to be “competitive”. What is important is that the market has a **contestable structure: low or nonexistent sunk costs, and the possibility to use strategies to undercut rivals, i.e., price adjustments by the firms are felt strongly by consumers.**

Perfect Competition

Price-taking behavior

In a **competitive market**, firms are price takers.

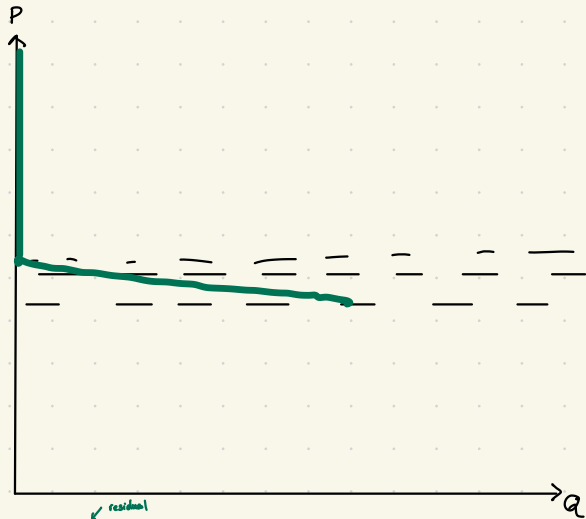
- In both input and output markets.
- A price-taking firm faces a demand curve that is **horizontal at the market price**.
 - The individual firm faces a perfectly elastic demand curve.

The Residual Demand Curve

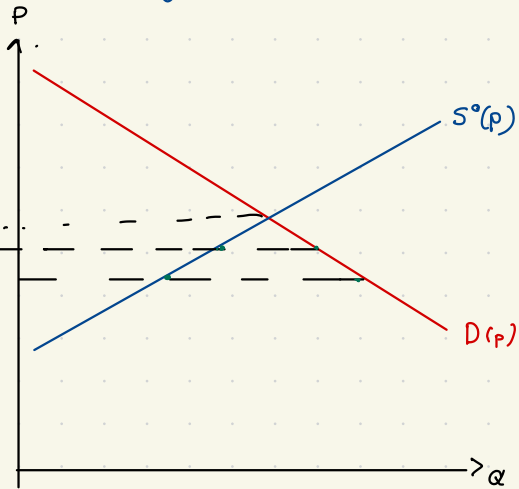
- Suppose Market demand is $D(p)$.  (Downward Sloping)
- Suppose that the (aggregate) supply curve of other firms is $S^0(p)$. 
- Then the residual demand curve for an individual firm is

$$D^r(p) = D(p) - S^0(p)$$

- What are the properties of $D^r(p)$?



S^o : all other firms



$\overset{\text{residual}}{\mathcal{E}_i^r} = n \mathcal{E}^D - (n-1) \mathcal{E}_0^S$
 $\hookrightarrow$ # of firms

$\uparrow n \Rightarrow \mathcal{E}_i^r$ will become very negative.

Revenues

Given a market demand curve $Q = D(p)$ with inverse demand curve $p = D^{-1}(Q)$, if the firm faces this demand curve (in full) revenues for the firm are defined as:

$$R(Q) = \text{Price} \times \text{Quantity} = D^{-1}(Q) \times Q.$$

For example, if $Q = a - bp$, the inverse demand is $p = a/b - Q/b$, thus

$$R(Q) = \left(\frac{a}{b} - \frac{Q}{b}\right) Q$$



Profit Maximization

The firm's profit function, for any q , is

$$\Pi(q) = R(q) - C(q)$$

← $C(q, \text{input prices})$

where $R(q)$ are revenues and $C(q)$ are costs as a function of quantity.

The firm's problem is then to **maximize profits** by choosing optimally the quantity offered.

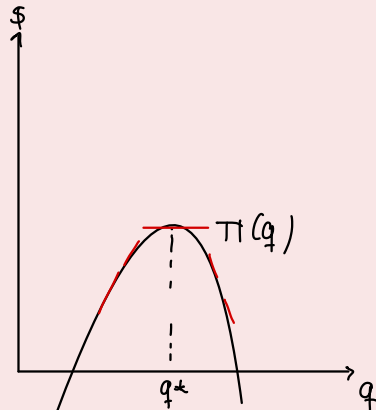
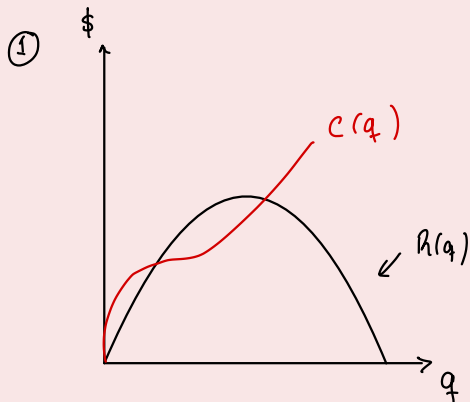
$$\max_q R(q) - C(q)$$

There are two steps in this problem:

- Output decision. What output level q^* maximizes profit or minimizes loss?
- Shutdown decision. Is it more profitable to produce q^* or to shutdown and produce no output?

Output rules

- 1 The firm sets its output where its profit is maximized.
- 2 The firm sets its output where marginal profit is equal to zero.
- 3 The firm sets its output where marginal revenue is equal to marginal cost.



$$\textcircled{2} \quad \pi(q) = R(q) - C(q) \quad \text{F.O.C.}$$

$$\frac{d\pi(q)}{dq} = \frac{dR(q)}{dq} - \frac{dC(q)}{dq} = 0$$

$$\downarrow \text{Marginal Profit}(q^*) = 0$$

$$\rightarrow MR(q^*) - MC(q^*) = 0$$



$$MR(q^*) = MC(q^*)$$

3

Shutdown rules

q^*

- 1 The firm shuts down only if it can reduce its loss by doing so.
- 2 The firm shuts down only if its revenue is less than its avoidable cost.

In the short-run: $C(q) = VC(q) + F$ Notice that $MC(q) = \frac{dVC(q)}{dq}$

$$\Pi(q) = R(q) - VC(q) - F$$

↳ Sunk costs (fixed costs) are irrelevant to the choice of q^*

↳ When the firm should shutdown?

→ If $R(q^*) < VC(q^*)$ then the firm should shutdown.

$$\hookrightarrow \Pi(0) = -F > \Pi(q^*) = \underbrace{R(q^*) - VC(q^*)}_{< 0} - F$$

→ If $R(q^*) \geq VC(q^*)$ then the firm should produce

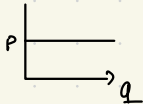
$$\hookrightarrow \Pi(q^*) = R(q^*) - VC(q^*) - F > \Pi(0) = -F$$

> 0

For a firm operating under perfect competition:

$$R(q) = p \cdot q$$

↳ Perfectly elastic demand curve



• If the firm decides to produce: $R(q^*) \geq VC(q^*)$, $q^* > 0$

$\Leftrightarrow$

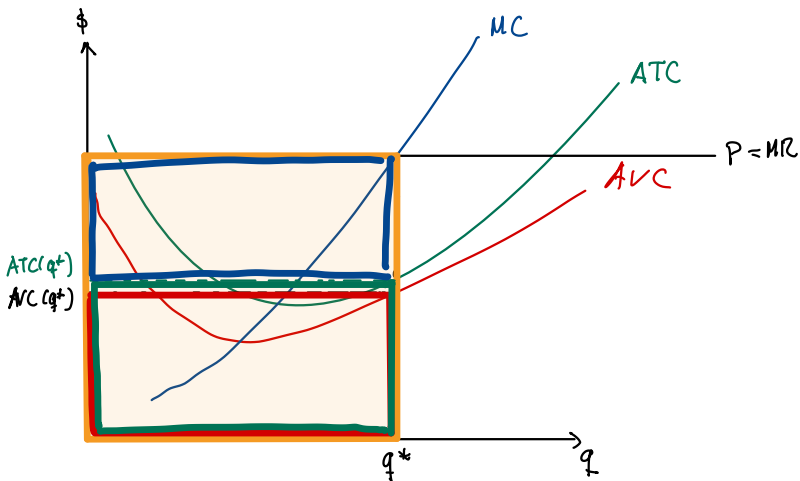
$$p \cdot q^* \geq VC(q^*)$$

("price for output needs to be greater or equal than the average variable cost")

$$p \geq \frac{VC(q^*)}{q^*} = AVC(q^*)$$

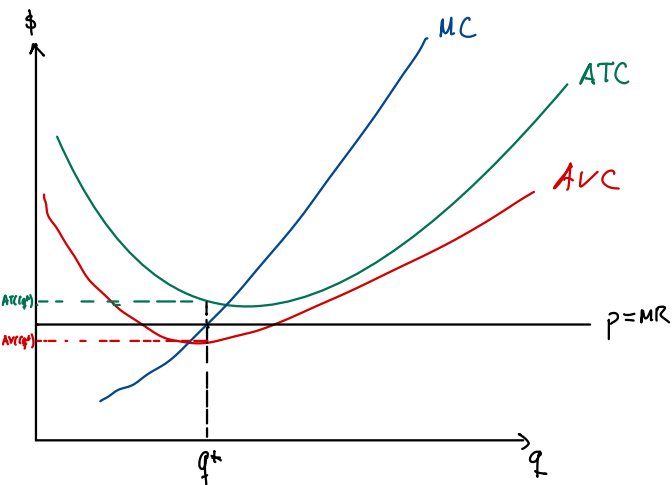
↳ Shutdown: Firm should shutdown if $p < AVC(q^*)$

Competition in the Short Run & the Short Run Output Decision



- $R(q^*)$
- $VC(q^*)$
- $TC(q^*)$
- Profits > 0

Competition in the Short Run & the Short Run Output Decision



$$R(q^*) = P q^*$$

$$VC(q^*) = AVC(q^*) q^*$$

$$TC(q^*) > R(q^*) > VC(q^*)$$

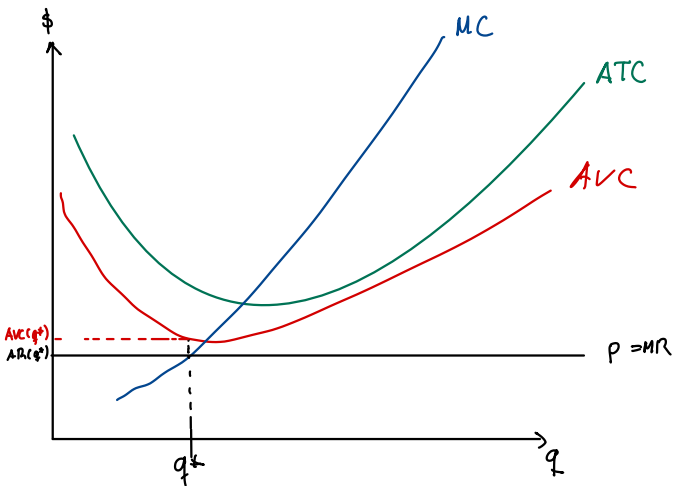
↳ firm should operate

Firm operates at a Loss

↳ in absolute terms:

$$TC(q^*) - R(q^*)$$

Competition in the Short Run & the Short Run Output Decision

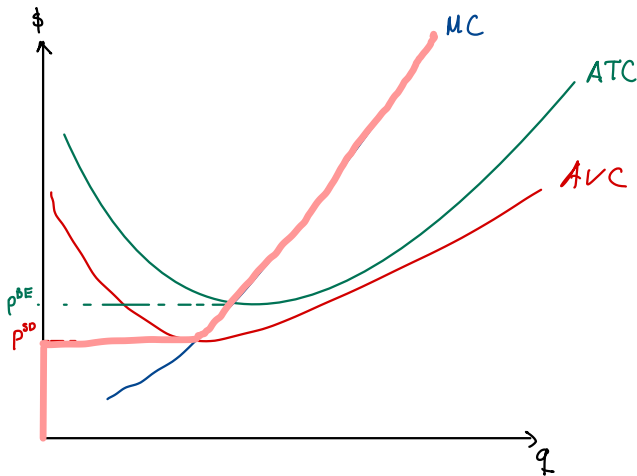


In this case the firm
should shut down because

$$R(q^*) < VC(q^*)$$

Competition in the Short Run & the Short Run Output Decision

What is the Supply Curve of the firm in the short-run?



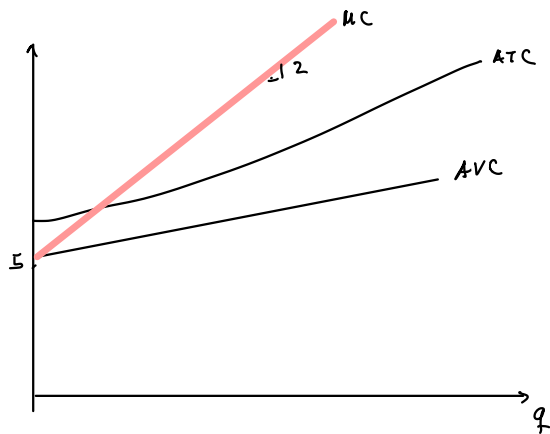
— Supply Curve
of the firm

$P = MC(q)$ as long
↓ as $P \geq P^{SD}$
(“inverse supply”)

↓ solve for q to get

$$q = S^{\text{individual}}(P) \quad P \geq P^{SD}$$

Competition in the Short Run & the Short Run Output Decision



$$MC(q) = 5 + 2q$$

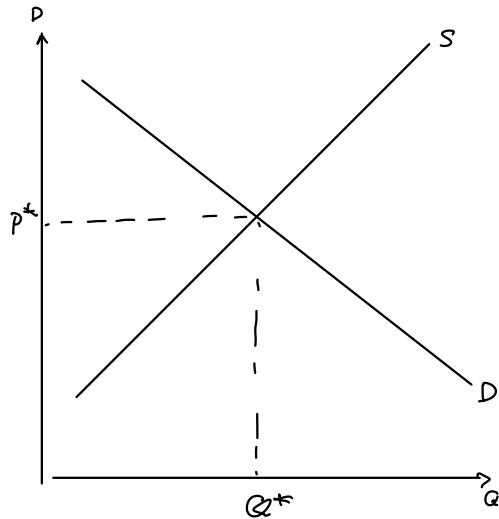
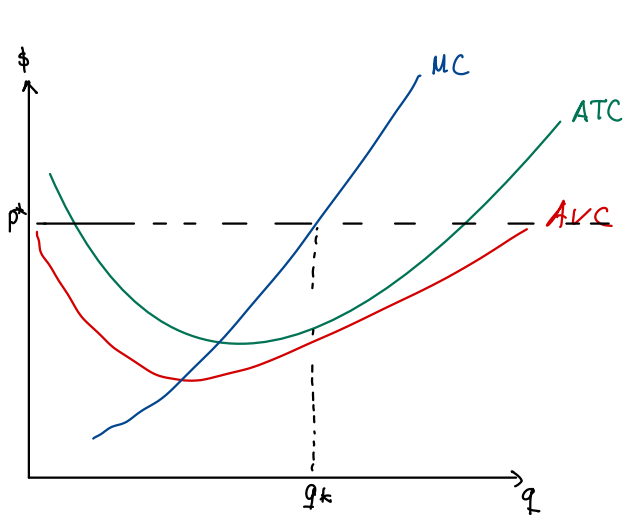
$$p = 5 + 2q \quad p \geq 5 \text{ (Shut-down price)}$$

$$q_s^i = \frac{p-5}{2} \quad (\text{individual supply curve})$$

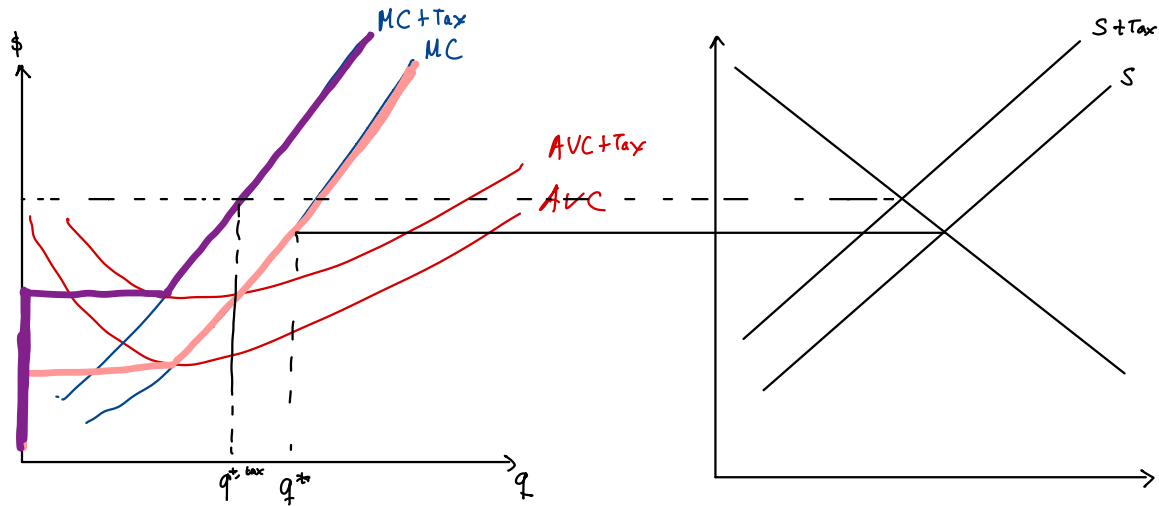
100 firms, identical

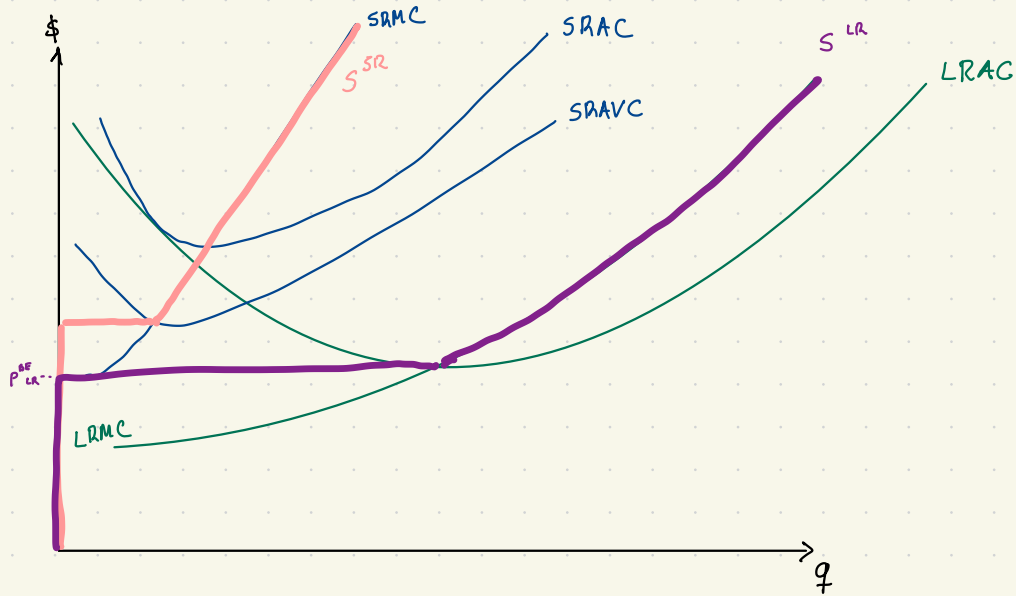
$$\begin{aligned} Q_s &= 100 q_s^i = 100 \left(\frac{p-5}{2} \right) \\ &= 50(p-5) \end{aligned}$$

Competition in the Short Run & the Short Run Output Decision

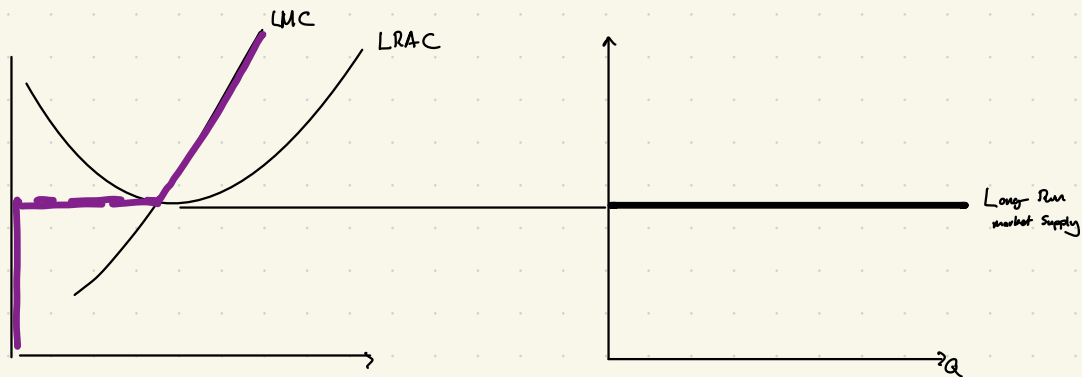


Competition in the Short Run & the Short Run Output Decision



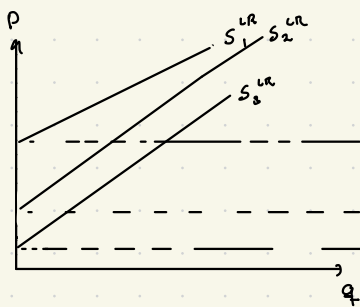


Identical firms with Free entry

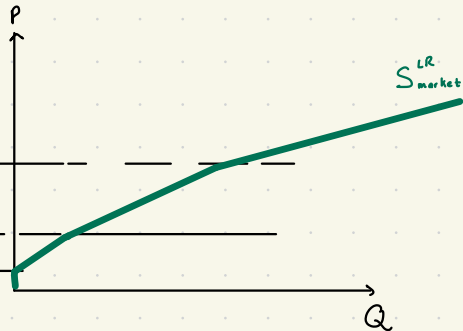


Upwards Sloping Long Run Market Supply?

- Non-Identical Firms



Horizontal
Aggregation $\rightarrow$



- Increasing Costs
- Limited Availability of Resources.